PRODUCING ALGEBRAIC INDEPENDENCE FROM A CODIMENSION TWO COMPONENT CONDITION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9011-9022 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9024-9052, the complete author proof (27 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Finitely generated domains over a field, dimension of vanishing loci, and algebraic independence in a rational function field; the argument is retained.

Source reference. Original lemma 0B2L: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9052–9052. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let A be a domain of finite type over a field k. Let $n \geq 2$. Let $g_1, \ldots, g_n \in A$ be elements such that $V(g_1, g_2)$ has an irreducible component of dimension $\dim(A) - 2$. Then there exist $c_1, \ldots, c_n \in k$ such that the elements*

$$x_1, \ldots, x_n, \sum g_i x_i, \sum c_i g_i \in \mathit{Frac}(A)(x_1, \ldots, x_n)$$

are algebraically independent over k.

Proof. The algebraic independence over k means that the morphism

$$T = \mathrm{Spec}(A[x_1, \ldots, x_n]) \longrightarrow \mathrm{Spec}(k[x_1, \ldots, x_n, y, z]) = S$$

given by $y = \sum g_i x_i$ and $z = \sum c_i g_i$ is dominant. Set $d = \dim(A)$. If $T \to S$ is not dominant, then the image has dimension $< n + 2$ and hence every irreducible component of every fibre has dimension $> d + n - (n + 2) = d - 2$, see Varieties, Lemma 0B2L. Choose a closed point $u \in V(g_1, g_2)$ contained in an irreducible component of dimension $d - 2$ and in no other component of $V(g_1, g_2)$. Consider the closed point $t = (u, 1, 0, \ldots 0)$ of T lying over u. Set $(c_1, \ldots, c_n) = (0, 1, 0, \ldots, 0)$. Then t maps to the point $s = (1, 0, \ldots, 0)$ of S. The fibre of $T \to S$ over s is cut out by

$$x_1 - 1, x_2, \ldots, x_n, \sum x_i g_i, g_2$$

and hence equivalently is cut out by

$$x_1 - 1, x_2, \ldots, x_n, g_1, g_2$$

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

By our condition on g_1, g_2 this subscheme has an irreducible component of dimension $d - 2$. $\square$